

This **Business Requirement Document (BRD)** is structured to provide clear, atomic requirements focused on the identified defects. Each requirement includes a specific **Navigation Path** for the application at <https://ecommbbs-stage.vercel.app/> and is designed to trigger a single, basic test scenario.

Business Requirement Document: E-Commerce Platform Optimization

1. Functional Requirements

1.1 Visual Search Accuracy

- **Requirement:** The image recognition engine must successfully map uploaded assets to active product categories.
- **Basic Test Scenario:** Upload an image of a shirt and verify that the system returns matching items.
- **Navigation Path:**
 1. Open <https://ecommbbs-stage.vercel.app/>
 2. Enable scan image toggle
 3. Click the **Camera/Scan Icon** in the Search bar.
 4. Upload the file `shirt.png`.
 5. Verify that the "Search Results" page is not empty.

1.2 Checkout Quantity Limits

- **Requirement:** The application must enforce a maximum quantity limit of 99 per item to prevent order processing errors.
- **Basic Test Scenario:** Increment an item's quantity to the limit and ensure it does not exceed 99.
- **Navigation Path:**
 1. Navigate to any Product Page.
 2. Select a size/color and click **"Add to Bag"**.
 3. Open the **Shopping Cart** (Side Drawer or Page).
 4. **Critical Step:** Proceed to the **Checkout Page**.
 5. Use the `+` button on the checkout summary to increase quantity.
 6. Verify the count stops and the button disables at 99.

1.3 Regional Promotional Display

- **Requirement:** Product listings must dynamically populate when a location-based promotional banner is active.
 - **Basic Test Scenario:** Access the Offers page from a valid region and verify product visibility.
 - **Navigation Path:**
 1. Click **"Offers"** in the top navigation menu.
 2. Ensure the banner displays: *"Special offers available for [Your Country]"*.
 3. Verify that the product grid below the banner displays at least one item.
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2. Visual & Interface Requirements

2.1 Product Grid Symmetry

- **Requirement:** All product display cards must maintain identical spacing and padding tokens.
- **Basic Test Scenario:** Verify that product cards in the "Featured" section are perfectly aligned.
- **Navigation Path:**
 1. Go to the Home Page.
 2. Scroll to the **"Featured Products"** grid.
 3. Verify that all card borders and images align horizontally and vertically without "jumping" gaps.

2.2 Tablet Responsive Layout

- **Requirement:** The checkout interface must adapt to medium-sized screens by maintaining a single-column layout for form fields.
- **Basic Test Scenario:** Inspect the checkout form on a tablet viewport.
- **Navigation Path:**
 1. Add an item to the cart and navigate to the **Checkout Page**.
 2. Resize the browser to a **Tablet width (e.g., 768px - 1024px)**.
 3. Verify the "First Name" and "Last Name" inputs are stacked vertically and do not split into separate columns.

2.3 Element Stacking (Z-Index)

- **Requirement:** Interactive navigation menus must always layer above background overlays.
- **Basic Test Scenario:** Trigger the mobile menu while an overlay is active.

- **Navigation Path:**
 1. Resize browser to **Mobile width**.
 2. Click the **Search Icon** to open the search overlay.
 3. Click the **Hamburger Menu (Three lines)**.
 4. Verify the menu appears clearly *on top* of the search overlay.

2.4 Text Readability

- **Requirement:** Global footer text must meet standard contrast levels for legibility.
 - **Basic Test Scenario:** Verify the visibility of footer links.
 - **Navigation Path:**
 1. Scroll to the very bottom of any page.
 2. Inspect the **Footer Section**.
 3. Verify that the light grey text is clearly readable against the background.
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3. Accessibility Requirements

3.1 Image Descriptions

- **Requirement:** All non-decorative product images must provide text alternatives for screen readers.
- **Basic Test Scenario:** Verify the presence of `alt` attributes on product thumbnails.
- **Navigation Path:**
 1. Right-click any product image on the Home Page.
 2. Select **"Inspect"**.
 3. Verify the `<img>` tag contains a descriptive `alt` attribute (not empty).

3.2 Form Focus Mapping

- **Requirement:** Form labels must be programmatically linked to their respective input fields.
- **Basic Test Scenario:** Click a text label to trigger input focus.
- **Navigation Path:**
 1. Navigate to the **Checkout Page**.
 2. Click directly on the text word **"First Name"**.
 3. Verify that the cursor automatically appears inside the corresponding input box.

3.3 Keyboard Interaction

- **Requirement:** All interactive UI controls must be navigable using the `Tab` key.
- **Basic Test Scenario:** Navigate the header using only a keyboard.

- **Navigation Path:**
 1. Click in the browser address bar, then press **Tab**.
 2. Continue pressing **Tab** to move through the navigation links and the **Cart Icon**.
 3. Verify that every interactive element shows a visible "focus ring" when selected.
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